
Financial Econometrics

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1. For the standard uniform random variable $X \sim Unif[0, 1]$ where the probability density function is $f(x) = 1$ for $0 \leq x \leq 1$ or otherwise calculate the mean, variance, skewness and kurtosis.

Mean: $E[X] = \int_0^1 1 \cdot x dx = [x^2/2]_0^1 = 1/2$

Variance: $\sigma^2 := E[X - E[X]]^2 = \int_0^1 (x - 1/2)^2 = [\frac{1}{3}(x - 1/2)^3]_0^1 = 1/3(1/2^3 - -1/2^3) = 2/3(1/8) = 1/12$

Skewness: $E[(X - E[X])/\sigma]^3 = 0$ since the distribution is symmetrical around mean $1/2$

Kurtosis: $E[((X - E[X])/\sigma)^4] = (\sqrt{12})^4 (\int_0^1 (x - 1/2)^4) = (12)^2 [1/5(x - 1/2)^5]_0^1 = (12)^2 * 1/5(1/2^5 + 1/2^5) = 144 * 2/5 * 1/64 = 144/160 = 9/10$

2. For two random variables X and Y , provide a proof of the Law of Iterated Expectations, namely that

$$E_Y[Y] = E_X[E_{Y|X}[Y|X]] \quad (1)$$

you may assume that X, Y are continuously distributed (likewise the proof can be shown for discrete variables).

$$E_X[E_{Y|X}[Y|X]] = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} y f(y|x) dy \right] f(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(y|x) f(x) dy dx$$

Note that by Bayes Rule $f(y|x)f(x) = f(y, x)$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(y|x) f(x) dy dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(y, x) dy dx = \int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f(y, x) dx \right) dy$$

Where by definition we get the marginal density of y by integrating the joint density function over all values x can take, i.e. $f(y) = \left(\int_{-\infty}^{\infty} f(y, x) dx \right)$

$$\int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f(y, x) dx \right) dy = \int_{-\infty}^{\infty} y f(y) dy = E_Y[Y]$$

3. The weekly returns of the Apple stock for 1/1/01 to 12/2/01 are

(0.101, 0.050, 0.135, 0.003, 0.0543, -0.072, -0.007, -0.007, -0.009).

Using this data set of size $T = 9$ answer the following (all the below can be done on a calculator, and I would advise you use a calculator in this example).

- (a) Calculate the sample mean and variance.

Sample Mean $\bar{x} = \frac{1}{T} \sum_{t=1}^T x_t$ in our case $T = 9$ and we substitute in the 9 samples values of returns $\bar{x} = 0.02849$

Sample Variance: $\frac{1}{T} \sum_{t=1}^T (x_t - \bar{x})^2 = \frac{1}{9} \sum_{t=1}^T (x_t - 0.02849)^2 = 0.398$

- (b) Calculate the sample autocorrelation at lag 1 and 2.

Sample correlation at lag k $\hat{\rho}(k) = \frac{1}{T-k} \sum_{t=k+1}^T (x_{t-k} - \bar{x})(x_t - \bar{x})$ which we evaluate at $T = 9$ for $k = 1$ and $k = 2$ noting we lose the first (the first two) observations for correlations at lag 1 and 2 respectively. Plug in $\bar{x}$ from above and the remaining elements from the data vector and sum (you need to be comfortable calculating these by hand for the class test and exam).

$\hat{\rho}(1) = 0.1178$ and $\hat{\rho}(2) = 0.586$

- (c) Perform a 95% two sided hypothesis test that the population correlation at lag 1 is zero (state all the steps, the test statistic and distribution and any assumptions made).
 Test stat: $\sqrt{9} * 0.1178 = 0.354$
 Under H_0 (under the stronger hypothesis of i.i.d data) then the test stat is drawn from a $N(0, 1)$ approximately (when T large). The 95% critical values for a two sided test are ± 1.96 and hence our test stat is within this range so there is insufficient evidence to reject the null hypothesis that the true population correlation at lag 1 is zero.
- (d) Perform a 95% two sided joint hypothesis test that the population correlation at lag 1 and 2 are both zero (state all the steps, the test statistic and distribution and any assumptions made).
 Q-Stat $\sum_{k=1}^2 (\sqrt{T} \hat{\rho}(k))^2 = 3.8183$. Under the null the test stat is approximately χ^2_2 distributed. The p-value for this test is 0.1482, and so we cannot find evidence to reject the null at the 5% level. (Alternatively you may find the 95% critical value from the statistical tables, in this case 5.991)
- (e) Why may we be sceptical of inference from the results in (c),(d).
4. Let $\mathcal{F}_t$ denote the information set of all past prices and $\mathcal{G}_t$ the information set with all past prices and all public information.
- (a) Provide a statement of the weak and semi strong form of the efficient market hypothesis (EMH) and explain this result.
 See the definition in Lecture 2.
- (b) Show that the semi strong form of EMH implies the weak form.
 Follows by the law of iterated expectations. $E[R_T | \mathcal{G}_t] = R_T^*$ for all $T > t$ and all public information sets $\mathcal{G}_t$. Since the $\mathcal{G}_T$ contains $\mathcal{F}_T$ then the average of R_T for all possible past prices is on average R_T^* in every possible world of public information- hence the average R_T conditional on $\mathcal{F}_t$ is R_T^* as no matter what the public info is the average of returns is the normal returns for all possible prices.
- (c) Show that the weak form EMH implies returns in different time periods are uncorrelated when the normal rate of return is constant.
 $R_T^* = R$ for all T (constant normal returns) hence WF EMH implies $E[R_T | \mathcal{F}_t] = R$ for all $T > t$ hence $E[R_T - R | \mathcal{F}_t] = 0$. Then take any $R_{T-j} - R$ for $j > 0$ then we can multiply both sides of $E[R_T - R | \mathcal{F}_t] = 0$ by $(R_{T-j} - R)$ and note that $T - j < T$ and hence R_{T-j} is known and fixed and so $E[(R_T - R)(R_{T-j} - R) | \mathcal{F}_t] = 0$. Then by law of iterated expectations $E[(R_T - R)(R_{T-j} - R)] = 0$ noting also $E[R_T] = R$ for all T by law of iterated expectations (namely take expectations on both sides of $E[R_T | \mathcal{F}_t] = R$). Hence $Cov(R_T, R_{T-j}) = E[(R_T - R)(R_{T-j} - R)] = 0$ for all $T, j > 0$ so that if the normal rate of return is constant EMH implies returns are uncorrelated in different time periods.
- (d) If normal rate of return is not constant, does the result in (c) hold, and if not explain why.
 If the normal rate of return isn't constant we have $E[R_T - R_T^* | \mathcal{F}_t] = 0$ and this implies $Cov(R_T - R_T^*, R_{T-j} - R_{T-j}^*) = 0$. In general $Cov(R_T - R_T^*, R_{T-j} - R_{T-j}^*)$ depends on the covariance of R_T^* and so $Cov(R_T, R_{T-j})$ does not equal $Cov(R_T - R_T^*, R_{T-j} - R_{T-j}^*)$.
5. Below is the joint distribution of returns R_t for $t = 1$ and $t = 2$ that take 3 values, 0, 1 or 2 percent.

(a) Derive the marginal distributions of R_1 and R_2 .

$$R_1: Pr(R_1 = 0) = 0.18 \quad Pr(R_1 = 1) = 0.41 \quad Pr(R_1 = 2) = 0.31$$

$$R_2: Pr(R_2 = 0) = 0.34 \quad Pr(R_2 = 1) = 0.36 \quad Pr(R_2 = 2) = 0.3$$

(b) Derive the mean and variance of R_1, R_2 .

$$\text{Mean of } R_1: E[R_1] = 0 * 0.18 + 1 * 0.41 + 2 * 0.31 = 1.03$$

$$\text{Mean of } R_2: E[R_2] = 0 * 0.34 + 1 * 0.36 + 2 * 0.3 = 0.96$$

(c) Find $Cov(R_1, R_2)$

$$Cov(R_1, R_2) = E[R_1 R_2] - E[R_1]E[R_2] = 1.11 - 0.96 * 1.13 = 0.0252 \text{ where}$$

$$E[R_1 R_2] = 1 * 1 * 0.11 + 1 * 2 * 0.15 + 2 * 1 * 0.19 + 2 * 2 * 0.08 = 1.11$$

(d) Find the conditional distribution of R_1 given $R_2 = 0, 1$ or 2 respectively.

$R_1 | R_2 = 0$: takes probability 0,1,2 with conditional prob $0.05/0.34 = 0.147$, $0.21/0.34 = 0.618$ and $0.08/0.34 = 0.235$ respectively (all using Bayes Rule)

$R_1 | R_2 = 1$ takes probability 0,1,2 with conditional prob $0.1/0.36 = 0.277$, $0.11/0.36 = 0.306$ and $0.15/0.36 = 0.417$ respectively

$R_1 | R_2 = 2$ takes probability 0,1,2 with conditional prob $0.03/0.3 = 0.09$, $0.19/0.3 = 0.63333$ and $0.08/0.3 = 0.26667$ respectively

(e) What does (d) tell us about the validity of the efficient market hypothesis?

We can see that R_1 and R_2 are dependent on each other as the outcomes of each impact the probabilities of

		R_2		
		0	1	2
R_1	0	0.05	0.1	0.03
	1	0.21	0.11	0.19
	2	0.08	0.15	0.08